

2.13. Vortex Sheets and Panels (Rough Draft)

While a 2-dimensional source sheet can be used to synthesize non-lifting flow over a 2-dimensional body of arbitrary cross-section, a source sheet produces no circulation. No potential flow that includes circulation can be synthesized from source sheets alone. In addition the Kutta-Joukowski law requires that, for any closed 2-dimensional body of arbitrary shape, the lift per unit span is always proportional to the circulation. Thus, no lifting potential flow can be synthesized from source sheets alone.

To aid in the synthesis of 2-dimensional lifting flows, we shall now consider one more elementary 2-dimensional potential flow. That is the vortex sheet. A 2-dimensional vortex sheet is formed from line-vortices in the same way that a 2-dimensional source sheet is formed from line-sources. An infinite number of infinitesimally weak line-vortices are combined in a side-by-side fashion as shown in Fig. 2.13.1. In this figure we are again looking at an edge view of the sheet with the line-vortices all perpendicular to the page. Again we let s be the distance measured along this vortex sheet as shown in Fig 2.13.1 and let $\gamma(s)$ be the vortex strength per unit length measured along s . The total strength associated with a differential segment of the vortex sheet located at s and having length ds is then $\gamma(s)ds$. Thus, the differential velocity induced by ds at an arbitrary point in space (x,y) , located a distance r from ds , is given by

$$d\mathbf{V} = -\frac{\gamma(s)}{2\pi r} \mathbf{i}_\theta ds \quad (2.13.1)$$

where $\mathbf{i}_\theta$ is the unit vector normal to a vector from ds to (x,y) , as shown in Fig. 2.13.1. The total velocity induced at the point (x,y) by the entire vortex sheet, extending from $s = 0$ to $s = l$, is then

$$\mathbf{V}(x,y) = -\frac{1}{2\pi} \int_{s=0}^l \frac{\gamma(s)}{r} \mathbf{i}_\theta ds \quad (2.13.2)$$

A vortex sheet with a completely arbitrary curved shape, such as that shown in Fig. 2.13.1, is required to exactly synthesize lifting flow over a 2-dimensional body of arbitrary shape. However, as was the case with the source sheet, a curved vortex sheet is often approximated using a series of planar segments. A planar vortex sheet is referred to as a vortex panel. This type of elementary flow is particularly useful for synthesizing flow around airfoils. Two-dimensional lifting flow over an airfoil section is commonly synthesized from a series of such vortex panels, placed around the surface of the airfoil section.

We shall now consider a differential segment of a vortex panel that lies on the x -axis at the location $x = x_o$ and has a length dx_o . The velocity induced at any point (x,y) by this differential vortex panel is normal to a line drawn between the point (x,y) and the point $(x_o,0)$ and has a magnitude inversely proportional to the distance between the two points. The x - and y -components of the velocity induced at the point (x,y) by this infinitesimally small section of the vortex panel are then given by

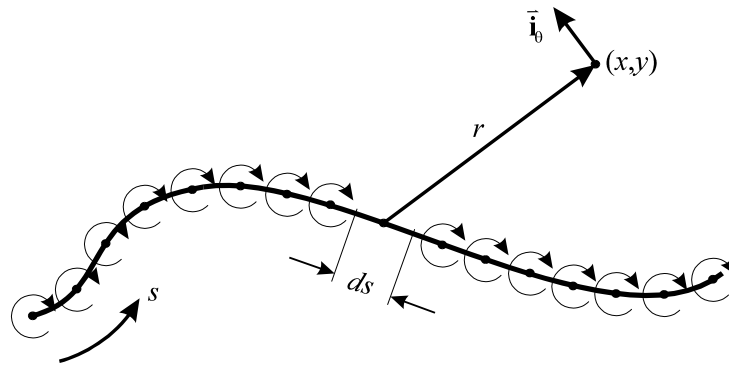


Figure 2.13.1. Edge view of a 2-dimensional vortex sheet.

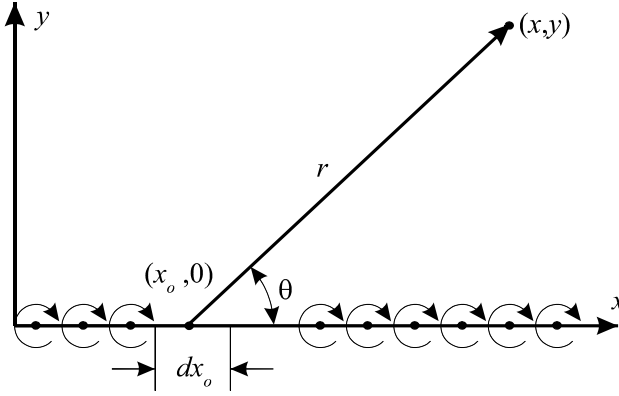


Figure 2.13.2. Edge view of a 2-dimensional vortex panel on the x -axis.

$$dV_x = -dV_\theta \sin \theta = \frac{\gamma(x_o)}{2\pi r} \sin \theta dx_o = \frac{y \gamma(x_o)}{2\pi [(x-x_o)^2 + y^2]} dx_o \quad (2.13.3)$$

and

$$dV_y = dV_\theta \cos \theta = -\frac{\gamma(x_o)}{2\pi r} \cos \theta dx_o = -\frac{(x-x_o) \gamma(x_o)}{2\pi [(x-x_o)^2 + y^2]} dx_o \quad (2.13.4)$$

where $\gamma(x_o)$ is the vortex strength per unit length at x_o and the remaining notation is defined in Fig. 2.13.2. For a vortex panel on the x -axis extending from $x = 0$ to $x = l$, we then have

$$V_x(x, y) = \frac{y}{2\pi} \int_{x_o=0}^l \frac{\gamma(x_o)}{(x-x_o)^2 + y^2} dx_o \quad (2.13.5)$$

$$V_y(x, y) = -\frac{1}{2\pi} \int_{x_o=0}^l \frac{(x-x_o) \gamma(x_o)}{(x-x_o)^2 + y^2} dx_o \quad (2.13.6)$$

These are exactly the same integrals that were encountered when working with the source panel. Thus, following the development from Eq. (2.12.14) through Eq. (2.12.20), a linear strength distribution with $\gamma(0) = \gamma_1$ and $\gamma(l) = \gamma_2$ will result in the x and y velocity components

$$\begin{aligned} V_x(x, y) &= \frac{y}{2\pi} \int_{x_o=0}^l \frac{\gamma_1 - \frac{x_o}{l}(\gamma_1 - \gamma_2)}{(x-x_o)^2 + y^2} dx_o \\ &= \frac{\gamma_1 l - x(\gamma_1 - \gamma_2)}{2\pi l} \tan^{-1} \left(\frac{yl}{y^2 + x^2 - xl} \right) + \frac{y(\gamma_1 - \gamma_2)}{4\pi l} \ln \left[\frac{x^2 + y^2}{(x-l)^2 + y^2} \right] \end{aligned} \quad (2.13.7)$$

$$\begin{aligned} V_y(x, y) &= -\frac{1}{2\pi} \int_{x_o=0}^l \frac{(x-x_o)[\gamma_1 - \frac{x_o}{l}(\gamma_1 - \gamma_2)]}{(x-x_o)^2 + y^2} dx_o \\ &= \frac{y(\gamma_1 - \gamma_2)}{2\pi l} \tan^{-1} \left(\frac{yl}{y^2 + x^2 - xl} \right) - \frac{\gamma_1 l - x(\gamma_1 - \gamma_2)}{4\pi l} \ln \left[\frac{x^2 + y^2}{(x-l)^2 + y^2} \right] - \frac{\gamma_1 - \gamma_2}{2\pi} \end{aligned} \quad (2.13.8)$$

Thus, after rearranging Eqs. (2.13.7) and (2.13.8), the velocity induced by a **2-dimensional vortex panel with linear strength distribution along the x-axis**, having length l with $\gamma(0) = \gamma_1$ and $\gamma(l) = \gamma_2$, is found to be

$$\begin{Bmatrix} V_x \\ V_y \end{Bmatrix} = \frac{1}{2\pi l} \begin{bmatrix} [(l-x)\Phi + y\Psi] & (x\Phi - y\Psi) \\ [y\Phi - (l-x)\Psi - l] & (-y\Phi - x\Psi + l) \end{bmatrix} \begin{Bmatrix} \gamma_1 \\ \gamma_2 \end{Bmatrix} \quad (2.13.9)$$

where

$$\Phi = \text{atan2}(yl, y^2 + x^2 - xl) \quad (2.13.10)$$

and

$$\Psi = \frac{1}{2} \ln \left[\frac{x^2 + y^2}{(x-l)^2 + y^2} \right] \quad (2.13.11)$$

The velocity along the upper and lower surfaces of the vortex panel is given by

$$V_x(x, \pm 0) = \pm \frac{\gamma_1 l - x(\gamma_1 - \gamma_2)}{2l} \quad (2.13.12)$$

$$V_y(x, \pm 0) = -\frac{\gamma_1 l - x(\gamma_1 - \gamma_2)}{4\pi l} \ln \left[\frac{x^2}{(x-l)^2} \right] - \frac{\gamma_1 - \gamma_2}{2\pi} \quad (2.13.13)$$

So the normal velocity is continuous across a vortex sheet while there is a step change in the tangential velocity. The change in tangential velocity across a vortex sheet is then given by the relation

$$V_x(x, +0) - V_x(x, -0) = \gamma_1 - \frac{x}{l}(\gamma_1 - \gamma_2) \quad (2.13.14)$$

or

$$V_x(x, +0) - V_x(x, -0) = \gamma(x) \quad (2.13.15)$$

Equation (2.13.15) points out an important relation that holds for all vortex sheets. At any point along a vortex sheet, the local jump in tangential velocity across the sheet is always equal to the local sheet strength. There is no normal velocity jump across a vortex sheet.